Chapter 29

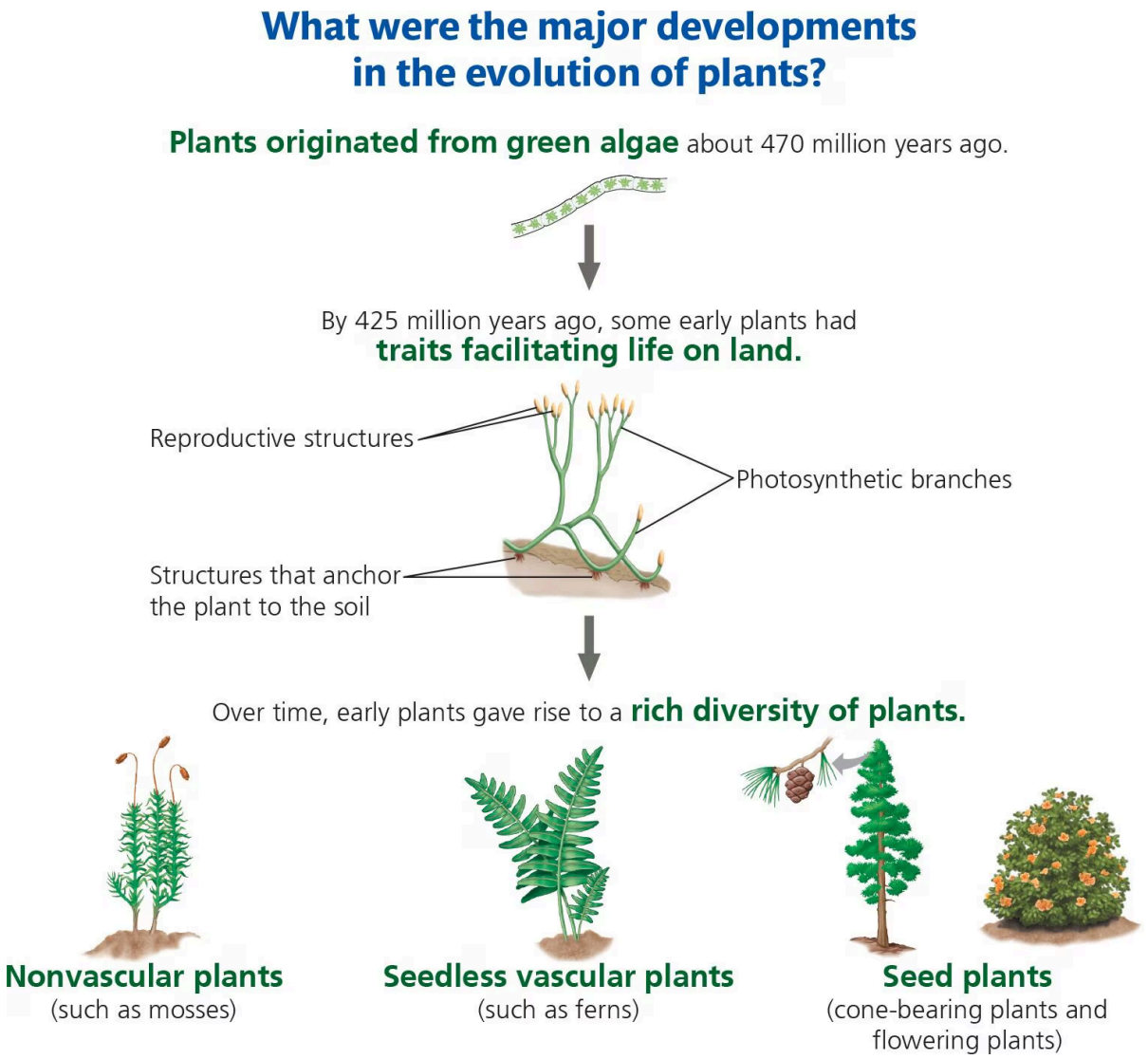
Campbell Biology, AP Edition, 12e

Chapter 29

Plant Diversity I: How Plants Colonized Land

Figure 29.1

For much of Earth's history, the land surface was largely lifeless. Some prokaryotes lived on land 3.2 billion years ago, but it was only within the last 500 million years that small plants, fungi, and animals joined them ashore. Finally, by 385 million years ago, the first forests appeared (but with different species than those of today).



Plants originated from green algae about 470 million years ago. By 425 million years ago, some early plants had traits facilitating life on land. These traits include reproductive structures, photosynthetic branches, and structures that anchor the plant to the soil. Over time, early plants gave rise to a rich diversity of plants, including the following: nonvascular plants (such as mosses), seedless vascular plants (such as ferns), and seed plants (such as cone-bearing plants and flowering plants).

Key Concepts

- 29.1**Plants evolved from green algae****
- 29.2**Mosses and other nonvascular plants have life cycles dominated by gametophytes****
- 29.3**Ferns and other seedless vascular plants were the first plants to grow tall****

As organisms colonized land, the changed environment acted as a selective mechanism for particular traits that ensured reproductive success **(BIG IDEA 1)** and led to differentiation in the use of energy as organisms exchanged matter with the environment **(BIG IDEA 2)** and interacted with each other, recycling matter **(BIG IDEA 4)**.

Study Tip

Make a table: As you read the chapter, make a table of adaptations for life on land that evolved in plants after they diverged from their close aquatic relatives, a group of charophyte algae.

Adaptation	Description	How it facilitates life on land
Cuticle	Waxy covering on outer body surface	Reduces dessication

The table is as follows:

Adaptation	Description	How it facilitates life on land
Cuticle	Waxy covering on outer body surface	Reduces desiccation
blank	blank	blank

Get Ready for This Chapter

In this chapter, you will learn about the colonization of land by plants, focusing on the first 100 million years of plant evolution. Before beginning this chapter, you should be able to do the following:

- Explain the process of photosynthesis (see Concept 10.2, Figure 10.5, and Figure 10.21).
- Define and apply the terms *haploid* and *diploid* (see Concept 13.2).
- Identify the algal group from which plants arose (see Figure 28.5).

The following questions provide a quick check of your basic knowledge in these areas.

1. Which of the following chemical equations summarizes the reactions in photosynthesis?

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Light energy converts carbon dioxide and water to sugars and oxygen.
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That's correct.

Correct! If you had trouble with this question, review the following material: Photosynthesis begins when light energy is used to split molecules of water. The process releases oxygen to the atmosphere and is used to convert light energy to the chemical energy of ATP and NADPH. Those molecules are then used (in the Calvin cycle) to convert carbon dioxide to sugars, such as glucose.

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Light energy converts oxygen and sugars to water and carbon dioxide.
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That's incorrect. Try again.

Incorrect. Review the following material: Photosynthesis begins when light energy is used to split molecules of water. The process releases oxygen to the atmosphere and is used to convert light energy to the chemical energy of ATP and NADPH. Those molecules are then used (in the Calvin cycle) to convert carbon dioxide to sugars, such as glucose.

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Sugars and oxygen convert to light energy, carbon dioxide, and water.
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That's incorrect. Try again.

Incorrect. Review the following material: Photosynthesis begins when light energy is used to split molecules of water. The process releases oxygen to the atmosphere and is used to convert light energy to the chemical energy of ATP and NADPH. Those molecules are then used (in the Calvin cycle) to convert carbon dioxide to sugars, such as glucose.

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Carbon dioxide and oxygen convert to light energy, sugars, and water.
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That's incorrect. Try again.

Incorrect. Review the following material: Photosynthesis begins when light energy is used to split molecules of water. The process releases oxygen to the atmosphere and is used to convert light energy to the chemical energy of ATP and NADPH. Those molecules are then used (in the Calvin cycle) to convert carbon dioxide to sugars, such as glucose.

2. If the diploid chromosome number of a sexually-reproducing species is 58, which of the following statements is true?

The haploid chromosome number is 116.

That's incorrect. Try again.

Incorrect. Review the following material: Each somatic cell of a sexually-reproducing species contains pairs of homologous chromosomes. One chromosome of each pair is inherited from each parent. For example, in humans, the 46 chromosomes in the somatic cells are actually two sets of 23 chromosomes—a maternal set (from the mother) and a paternal set (from the father). The number of chromosomes in a single set is represented by n. Any cell with two chromosome sets has a diploid number of chromosomes, abbreviated 2n. For humans, the diploid number is 46 (2n=46). Unlike somatic cells, gametes contain a single set of chromosomes. Such cells have a haploid number of chromosomes (n). For humans, the haploid number is 23 (n=23).

Each gamete contains 58 chromosomes.

That's incorrect. Try again.

Incorrect. Review the following material: Each somatic cell of a sexually-reproducing species contains pairs of homologous chromosomes. One chromosome of each pair is inherited from each parent. For example, in humans, the 46 chromosomes in the somatic cells are actually two sets of 23 chromosomes—a maternal set (from the mother) and a paternal set (from the father). The number of chromosomes in a single set is represented by n. Any cell with two chromosome sets has a diploid number of chromosomes, abbreviated 2n.

For humans, the diploid number is 46 (2n=46). Unlike somatic cells, gametes contain a single set of chromosomes. Such cells have a haploid number of chromosomes (n). For humans, the haploid number is 23 (n=23).

There are 58 pairs of homologous chromosomes in each cell.

That's incorrect. Try again.

Incorrect. Review the following material: Each somatic cell of a sexually-reproducing species contains pairs of homologous chromosomes. One chromosome of each pair is inherited from each parent. For example, in humans, the 46 chromosomes in the somatic cells are actually two sets of 23 chromosomes—a maternal set (from the mother) and a paternal set (from the father). The number of chromosomes in a single set is represented by n. Any cell with two chromosome sets has a diploid number of chromosomes, abbreviated 2n. For humans, the diploid number is 46(2n=46). Unlike somatic cells, gametes contain a single set of chromosomes. Such cells have a haploid number of chromosomes (n). For humans, the haploid number is 23 (n=23).

The haploid chromosome number is 29.

That's correct.

Correct! If you had trouble with this question, review the following material: Each somatic cell of a sexually-reproducing species contains pairs of homologous chromosomes. One chromosome of each pair is inherited from each parent. For example, in humans, the 46 chromosomes in the somatic cells are actually two sets of 23 chromosomes—a maternal set (from the mother) and a paternal set (from the father). The number of chromosomes in a single set is represented by n. Any cell with two chromosome sets has a diploid number of chromosomes, abbreviated 2n. For humans, the diploid number is 46(2n=46). Unlike somatic cells, gametes contain a single set of chromosomes. Such cells have a haploid number of chromosomes (n). For humans, the haploid number is 23(n=23).

3. Plants are members of the eukaryotic supergroup _____, and their closest relatives are _____.

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Unikonta; charophyte green algae.
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That's incorrect. Try again.

Incorrect. Review the following material: There are four eukaryotic supergroups: Excavata, SAR, Archaeplastida, and Unikonta. Plants are members of Archaeplastida. Charophyte green algae are the sister group to plants; thus, the closest relatives of plants are charophyte green algae.

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Archaeplastida; charophyte green algae.
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That's correct.

Correct! If you had trouble with this question, review the following material: There are four eukaryotic supergroups: Excavata, SAR, Archaeplastida, and Unikonta. Plants are members of Archaeplastida. Charophyte green algae are the sister group to plants; thus, the closest relatives of plants are charophyte green algae.

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Archaeplastida; chlorophyte green algae.
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That's incorrect. Try again.

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Learning Objectives

After studying this chapter, you should be able to do the following:

- 29.1Identify key derived characters of plants.
- 29.2Characterize the life cycles of nonvascular plants.
- 29.3Describe characteristics and reproductive processes of seedless vascular plants.